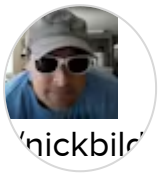
I Heart Wi-Fi

UCSC's Pulse-Fi uses Wi-Fi signals from ESP32 chips and AI to measure heart rate with clinical accuracy from up to 10 feet away.



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This ESP32 is measuring heart rate with Wi-Fi and AI (: Erika Cardema / UC Santa Cruz)

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Wi-Fi has had quite an exciting few months! After decades of us thinking it was all about wireless networking, it turns out it has a few other tricks up its sleeve. First we found out that Wi-Fi knows exactly who you are (https://www.hackster.io/news/i-m-sensing-a-disturbance-in-the-wi-fi-5c3249aabd93), and now it has even learned to measure your heart rate. I can hardly wait to see what Wi-Fi is going to do next. The way things are going, it will probably be something between reading our minds and taking over the world.

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Traditionally, heart rate has been measured by a wearable monitor of some sort. But now researchers at the University of California, Santa Cruz have demonstrated that no physical contact is necessary. By leveraging Wi-Fi signals generated by ESP32 chips and Raspberry Pi computers, the team has come up with a system that can measure a person's heart rate (https://news.ucsc.edu/2025/09/pulse-fi-wifi-heart-rate/) from up to 10 feet away.

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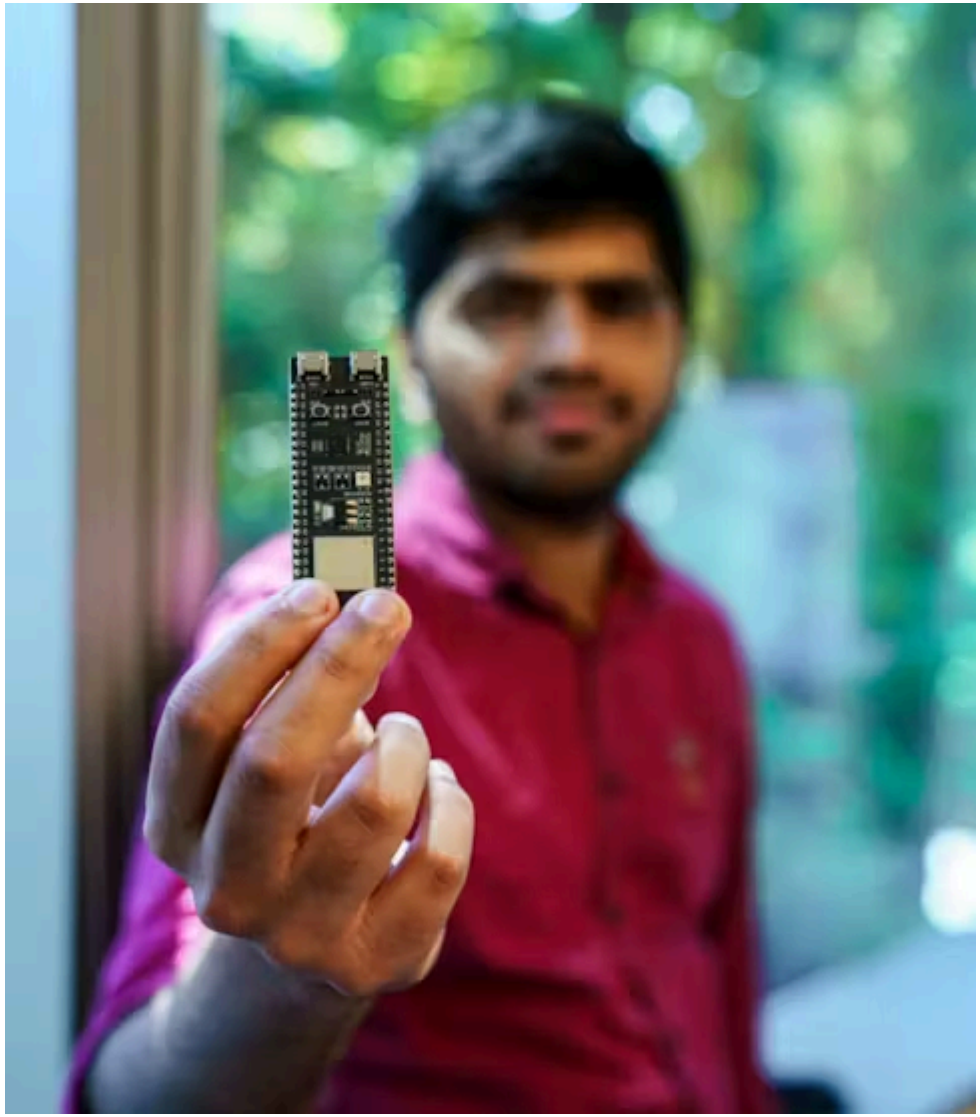

An ESP32 development board used in the study (📷 : Erika Cardema / UC Santa Cruz)

The technology, called Pulse-Fi, uses Wi-Fi radio waves paired with machine learning algorithms to detect subtle changes in the signal caused by a beating heart. The results show that after just five seconds of monitoring, the system

can achieve clinical-level accuracy, with an error margin of only half a beat per minute.

So how does it work? Wi-Fi devices constantly emit radio frequency waves, which interact with the physical environment as they travel to a receiver. When these waves pass through or bounce off objects (like people) they undergo slight alterations. Pulse-Fi's algorithm was trained to pick out the faint variations specifically linked to heartbeats, while filtering out noise from environmental movement or other signal disruptions. This sensitivity to tiny fluctuations is what allows the system to capture heart rate data without requiring skin contact or other specialized equipment.

To validate the system, the team ran tests on 118 participants across 17 different body positions. Whether subjects were sitting, standing, lying down, or even walking, the results remained consistent. And thanks to the use of inexpensive hardware — ESP32 modules that cost as little as $5, or Raspberry Pi boards in the neighborhood of $30 — the technology has the potential to make non-intrusive health monitoring accessible even in resource-limited settings. The researchers note that more advanced Wi-Fi hardware, such as commercial routers, could improve accuracy even further.

Members of the research team (📷 : Erika Cardema / UC Santa Cruz)

Due to a lack of existing data in the area, the researchers had to create their own training data for their model. Using their custom ESP32 setup alongside standard oximeters, they generated a dataset that was used to teach their neural network how to recognize the telltale patterns of a heartbeat.

In the future, the group is planning to extend the system to track breathing rates as well, with early results showing promise for detecting conditions like sleep apnea. If successful, Pulse-Fi could evolve into a full suite of non-intrusive, low-cost health monitoring tools powered by nothing more invasive than the Wi-Fi signals already filling our homes.

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